1-4

October 15, 2026

[14]:
```
#Research Question
#Does mean age differ significantly across the five levels of AI familiarity?

#Research Hypotheses
#Null Hypothesis (H ):
#There is no statistically significant difference in mean age across the five␣
 ↪AI familiarity groups.
#[H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5]

#Alternative Hypothesis (H ):
#At least one AI familiarity group has a different mean age from another group.
#[H_1: \text{Not all five group means are equal}]

import pandas as pd

df = pd.read_excel(
    "PYTHON .xlsx",
)

print(df.head())
print(df.shape)
print(df.info())
```

```
         w5name  age  w5dq4
0  CRON3w5e01_1   69      1
1  CRON3w5e01_1   76      4
2  CRON3w5e01_1   44      2
3  CRON3w5e01_1   58      1
4  CRON3w5e01_1   38      5
(9544, 3)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9544 entries, 0 to 9543
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   w5name  9544 non-null   object
 1   age     9544 non-null   int64
 2   w5dq4   9544 non-null   int64
```

```
dtypes: int64(2), object(1)
memory usage: 223.8+ KB
None
```

[15]:
```python
# Make a copy of the original data
clean_df = df.copy()

# Remove records where age is 999
clean_df = clean_df[clean_df["age"] != 999]

# Remove records where familiarity is 99 or 6
clean_df = clean_df[(clean_df["w5dq4"] != 99) & (clean_df["w5dq4"] != 6)]

# Check the cleaning results
print("Original number of rows:", len(df))
print("Number of rows after cleaning:", len(clean_df))
print("Number of rows removed:", len(df) - len(clean_df))
print("Age = 999:", (clean_df["age"] == 999).sum())
print("Familiarity = 99:", (clean_df["w5dq4"] == 99).sum())
print("Familiarity = 6:", (clean_df["w5dq4"] == 6).sum())

#clean_df.to_excel(
#    "cleaned_data.xlsx",
#    index=False
#)
```

```
Original number of rows: 9544
Number of rows after cleaning: 8101
Number of rows removed: 1443
Age = 999: 0
Familiarity = 99: 0
Familiarity = 6: 0
```

[16]:
```python
# Calculate descriptive statistics for each AI familiarity group

group_stats = clean_df.groupby("w5dq4")["age"].agg(
    ["count", "mean", "median", "var", "std", "min", "max"]
)

print(group_stats.round(2))


# Check the AI familiarity groups
#print("AI familiarity groups:")
#print(sorted(clean_df["w5dq4"].unique()))

# Check the total number of observations
```

```
#print("Total observations:", clean_df["age"].count())

# Check the number of observations in each group
#print(clean_df["w5dq4"].value_counts().sort_index())

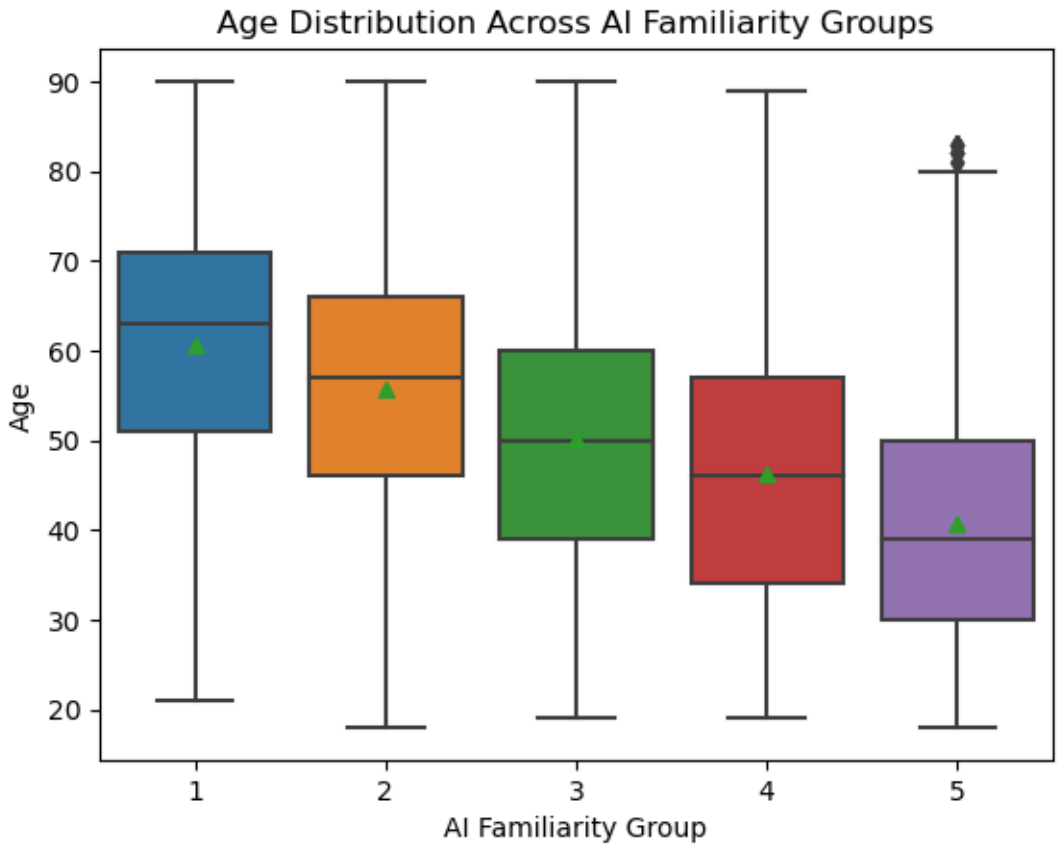
# Check whether group counts add up to the total
#print("Sum of group counts:", group_stats["count"].sum())
```

w5dq4	count	mean	median	var	std	min	max
1	2660	60.53	63.0	197.54	14.05	21	90
2	1390	55.60	57.0	204.86	14.31	18	90
3	1385	50.19	50.0	223.97	14.97	19	90
4	1223	46.37	46.0	231.50	15.22	19	89
5	1443	40.66	39.0	190.11	13.79	18	83

[18]:
```
import matplotlib.pyplot as plt
import seaborn as sns

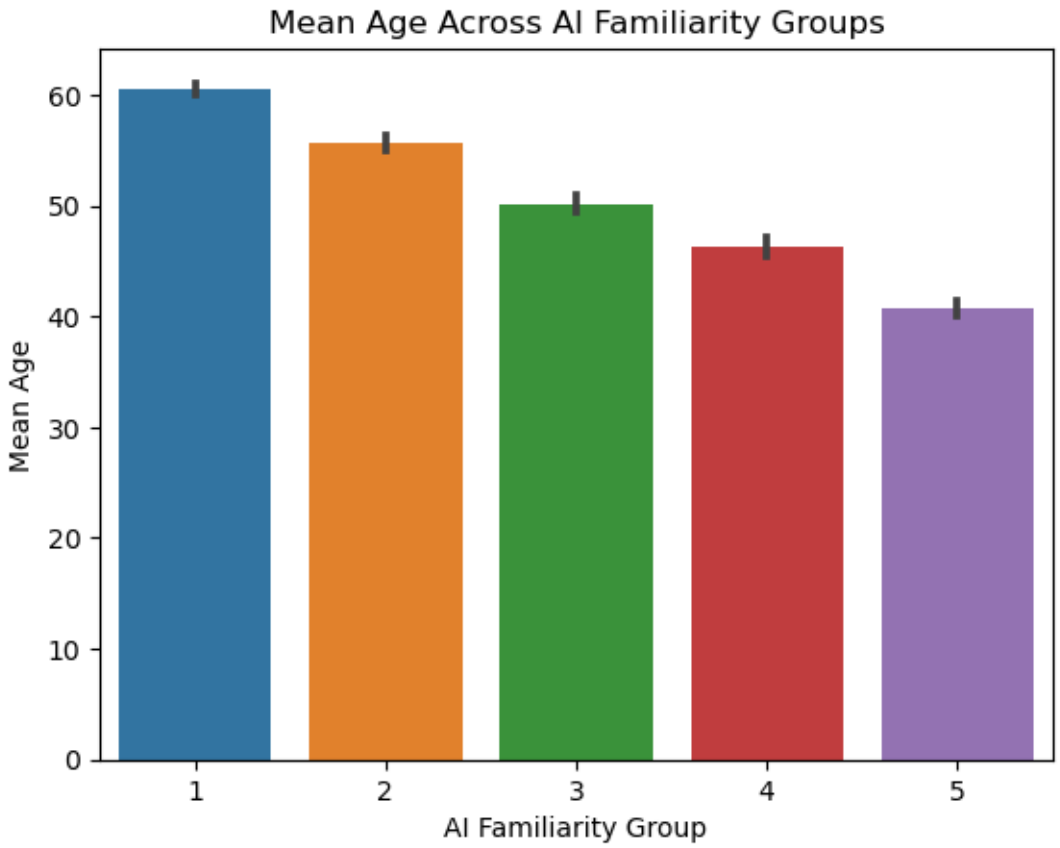
sns.boxplot(
    data=clean_df,
    x="w5dq4",
    y="age",
    showmeans=True
)

plt.xlabel("AI Familiarity Group")
plt.ylabel("Age")
plt.title("Age Distribution Across AI Familiarity Groups")
plt.show()
```



[22]:
```
sns.barplot(data=clean_df, x="w5dq4", y="age",
            #errorbar=None,
            errorbar=("ci", 95))

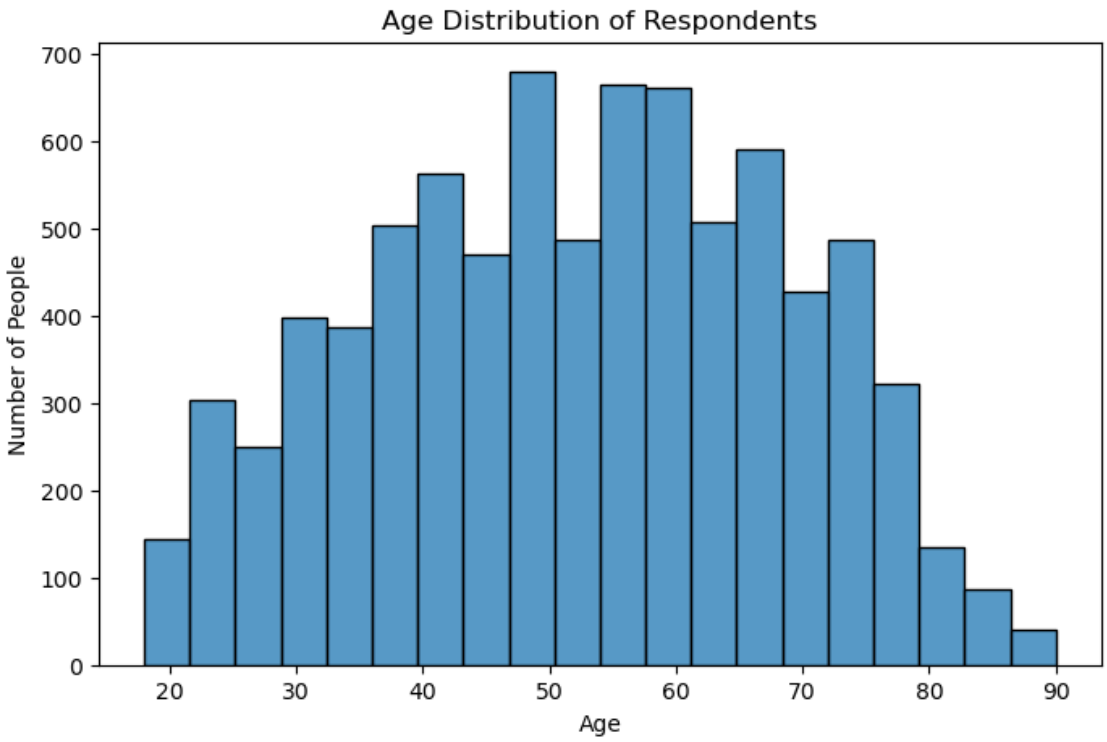
plt.xlabel("AI Familiarity Group")
plt.ylabel("Mean Age")
plt.title("Mean Age Across AI Familiarity Groups")
plt.show()
```



[20]:
```
plt.figure(figsize=(8, 5))

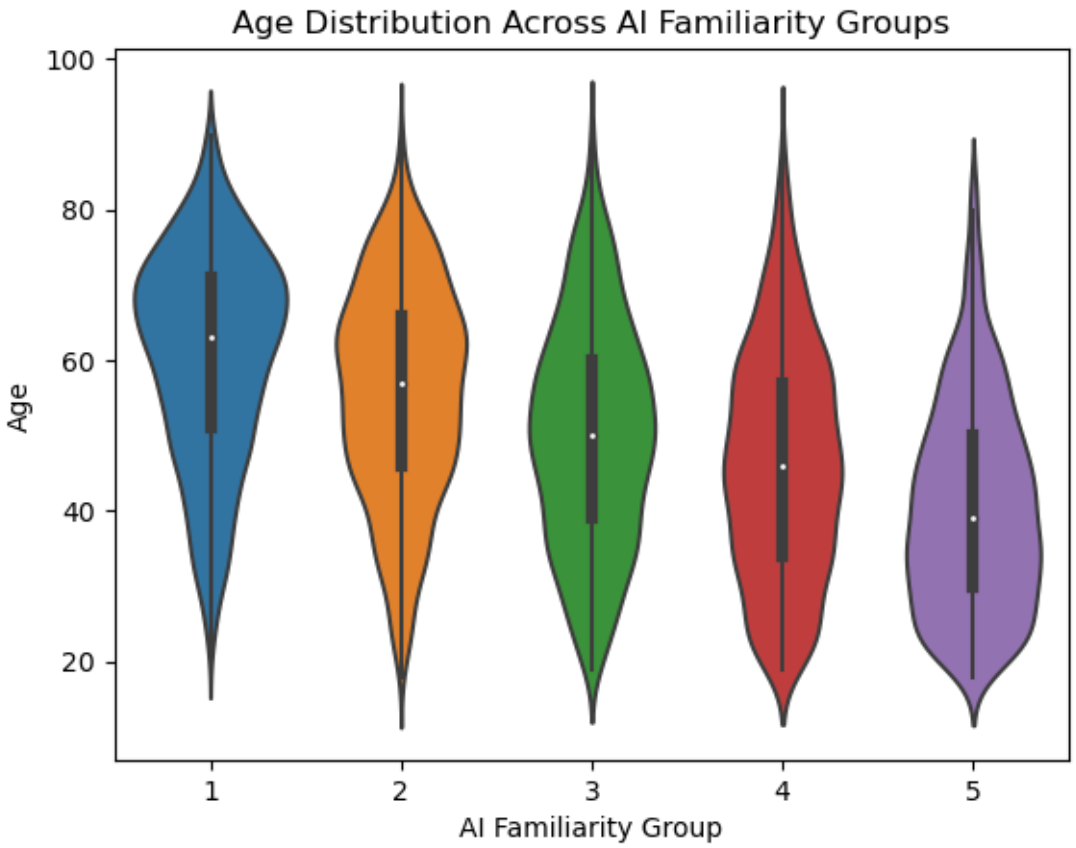
sns.histplot(data=clean_df, x="age", bins=20)

plt.xlabel("Age")
plt.ylabel("Number of People")
plt.title("Age Distribution of Respondents")
plt.show()
```



[21]:
```
sns.violinplot(data=clean_df, x="w5dq4", y="age")

plt.xlabel("AI Familiarity Group")
plt.ylabel("Age")
plt.title("Age Distribution Across AI Familiarity Groups")
plt.show()
```


Age Distribution Across AI Familiarity Groups
100
80
60
40
20
Age
1
2
3
4
5
AI Familiarity Group

[ ]: